

Electricity

MCQ

1. Electricity is a form of

- a) Matter
- b) Energy
- c) Force
- d) Substance

Answer: b) Energy

2. Electricity is mainly used to power

- a) Rivers
- b) Mountains
- c) Homes and appliances
- d) Forests

Answer: c) Homes and appliances

3. Electricity helps us understand how

- a) Plants grow
- b) Machines work
- c) Animals move
- d) Weather changes

Answer: b) Machines work

4. Who discovered electricity through a kite experiment?

- a) Thomas Edison
- b) Isaac Newton
- c) Benjamin Franklin
- d) Galileo

Answer: c) Benjamin Franklin

5. Benjamin Franklin's kite experiment showed similarity between static electricity and

- a) Rain
- b) Wind
- c) Thunder
- d) Light

Answer: c) Thunder

6. In which year did Edison light up New York City?

- a) 1700
- b) 1800
- c) 1882
- d) 1899

Answer: c) 1882

7. Electricity first came to India in

- a) 1850
- b) 1882
- c) 1899
- d) 1905

Answer: c) 1899

8. The first thermal power plant in India was commissioned at

- a) Delhi
- b) Mumbai
- c) Calcutta
- d) Chennai

Answer: c) Calcutta

9. The centre of an atom is called

- a) Orbit
- b) Electron
- c) Nucleus
- d) Shell

Answer: c) Nucleus

10. Which particle has no charge?

- a) Proton
- b) Electron
- c) Neutron
- d) Ion

Answer: c) Neutron

11. Protons are

- a) Negatively charged
- b) Positively charged
- c) Neutral
- d) Variable

Answer: b) Positively charged

12. Electrons revolve around the nucleus in

- a) Straight lines
- b) Circular orbits
- c) Random paths
- d) Waves

Answer: b) Circular orbits

13. Electricity is associated with

- a) Heat

- b) Light
- c) Electric charges
- d) Sound

Answer: c) Electric charges

14. Electric charge is denoted by

- a) I
- b) V
- c) Q
- d) R

Answer: c) Q

15. The unit of electric charge is

- a) Ampere
- b) Volt
- c) Ohm
- d) Coulomb

Answer: d) Coulomb

16. One coulomb is the charge of approximately

- a) 10^2 electrons
- b) 10^6 electrons
- c) 6.242×10^{28} electrons
- d) 10^{12} electrons

Answer: c) 6.242×10^{28} electrons

17. Electricity is defined as the movement of

- a) Protons
- b) Neutrons
- c) Electrons

d) Atoms

Answer: c) Electrons

18. Electric current flows only when there is

a) Heat

b) Resistance

c) Potential difference

d) Magnet

Answer: c) Potential difference

19. Conventional current is the flow of

a) Electrons

b) Neutrons

c) Positive charges

d) Atoms

Answer: c) Positive charges

20. Conventional current flows from

a) Lower to higher potential

b) Higher to lower potential

c) Left to right

d) Right to left

Answer: b) Higher to lower potential

21. Actual electron flow is from

a) Higher to lower potential

b) Lower to higher potential

c) North to south

d) South to north

Answer: b) Lower to higher potential

22. Electric current is the amount of charge flowing per

- a) Distance
- b) Area
- c) Unit time
- d) Temperature

Answer: c) Unit time

23. The SI unit of electric current is

- a) Volt
- b) Ohm
- c) Ampere
- d) Coulomb

Answer: c) Ampere

24. One ampere means

- a) One volt per second
- b) One coulomb per second
- c) One ohm per second
- d) One joule per second

Answer: b) One coulomb per second

25. Electric current is measured using

- a) Voltmeter
- b) Ammeter
- c) Galvanometer
- d) Thermometer

Answer: b) Ammeter

26. An ammeter is always connected in

- a) Parallel

- b) Series
- c) Diagonal
- d) Open circuit

Answer: b) Series

27. Smaller currents are measured using

- a) Multimeters
- b) Micro ammeters
- c) Voltmeters
- d) Batteries

Answer: b) Micro ammeters

28. Potential difference is also known as

- a) Current
- b) Resistance
- c) Voltage
- d) Power

Answer: c) Voltage

29. The unit of potential difference is

- a) Ampere
- b) Volt
- c) Ohm
- d) Coulomb

Answer: b) Volt

30. Potential difference is measured using

- a) Ammeter
- b) Voltmeter
- c) Thermometer

d) Barometer

Answer: b) Voltmeter

31. Charges move like water flowing from

a) Low level to high level

b) High level to low level

c) Same level

d) Random level

Answer: b) High level to low level

32. Resistance is the opposition to

a) Voltage

b) Power

c) Electric current

d) Heat

Answer: c) Electric current

33. The SI unit of resistance is

a) Ampere

b) Volt

c) Ohm

d) Watt

Answer: c) Ohm

34. Electrical resistivity is represented by

a) R

b) ρ

c) V

d) I

Answer: b) ρ

35.The SI unit of resistivity is

- a) Ohm
- b) Ohm metre
- c) Volt
- d) Ampere

Answer: b) Ohm metre

36.Electrical conductivity is the ability to

- a) Resist current
- b) Produce heat
- c) Conduct current
- d) Store charge

Answer: c) Conduct current

37.The unit of conductivity is

- a) Ohm
- b) Volt
- c) Siemens
- d) Ampere

Answer: c) Siemens

38.Conductors generally have

- a) High resistance
- b) High resistivity
- c) Low resistance
- d) No electrons

Answer: c) Low resistance

39.Water flow is compared to electric current to explain

- a) Pressure

b) Circuit behaviour

c) Resistance only

d) Voltage only

Answer: b) Circuit behaviour

40. Energy loss in a circuit appears as

a) Light

b) Sound

c) Heat

d) Motion

Answer: c) Heat

41. An electric cell converts

a) Electrical energy to heat

b) Chemical energy to electrical energy

c) Heat energy to light

d) Sound energy to electrical

Answer: b) Chemical energy to electrical energy

42. An electric cell has

a) One terminal

b) Two terminals

c) Three terminals

d) No terminals

Answer: b) Two terminals

43. Cells are classified based on

a) Size

b) Shape

c) Continuity of current

d) Colour

Answer: c) Continuity of current

44. A primary cell can be used

a) Many times

b) Only once

c) Forever

d) After recharging

Answer: b) Only once

45. A secondary cell can be

a) Used once

b) Recharged

c) Thrown away

d) Broken

Answer: b) Recharged

46. A dry cell was developed by

a) Edison

b) Volta

c) Yei Sakizo

d) Franklin

Answer: c) Yei Sakizo

47. Electrolyte helps in

a) Insulation

b) Ion transport

c) Heating

d) Cooling

Answer: b) Ion transport

48. In a dry cell, zinc chloride is used to

- a) Produce electricity
- b) Maintain moistness
- c) Act as electrode
- d) Prevent leakage

Answer: b) Maintain moistness

49. Carbon rod in a dry cell acts as

- a) Anode
- b) Cathode
- c) Electrolyte
- d) Insulator

Answer: b) Cathode

50. Manganese dioxide acts as a

- a) Conductor
- b) Electrolyte
- c) Depolarizer
- d) Insulator

Answer: c) Depolarizer

51. Dry cells are commonly used in

- a) Refrigerators
- b) Wall clocks
- c) Washing machines
- d) Fans

Answer: b) Wall clocks

52. A battery is a combination of

- a) Wires

b) Switches

c) Cells

d) Bulbs

Answer: c) Cells

53.The inventor of the first battery was

a) Galvani

b) Edison

c) Volta

d) Faraday

Answer: c) Volta

54.The first battery was called

a) Dry cell

b) Power cell

c) Voltaic pile

d) Primary cell

Answer: c) Voltaic pile

55.An electric switch is used to

a) Store energy

b) Break or close a circuit

c) Increase voltage

d) Reduce resistance

Answer: b) Break or close a circuit

56.When a switch is OFF, the circuit is

a) Closed

b) Complete

c) Open

d) Shorted

Answer: c) Open

57. An electric circuit must be

a) Open

b) Broken

c) Closed

d) Curved

Answer: c) Closed

58. In a series circuit, current flows through

a) Many paths

b) Two paths

c) A single path

d) No path

Answer: c) A single path

59. In a series circuit, if one element fails

a) Others work

b) Whole circuit stops

c) Voltage increases

d) Current increases

Answer: b) Whole circuit stops

60. In a parallel circuit, current has

a) One path

b) No path

c) Two or more paths

d) Circular path

Answer: c) Two or more paths

61. In a parallel circuit, if one element fails

- a) Entire circuit fails
- b) Battery explodes
- c) Other elements work
- d) Current stops everywhere

Answer: c) Other elements work

62. Conductors allow charges to pass

- a) With difficulty
- b) Easily
- c) Slowly
- d) Never

Answer: b) Easily

63. Insulators have

- a) High conductivity
- b) Low resistivity
- c) High resistivity
- d) Free electrons

Answer: c) High resistivity

64. Semiconductors have conductivity

- a) Higher than conductors
- b) Lower than insulators
- c) Between conductors and insulators
- d) Zero

Answer: c) Between conductors and insulators

65. Silicon and germanium are

- a) Conductors

- b) Insulators
- c) Semiconductors
- d) Electrolytes

Answer: c) Semiconductors

66. Heating effect of current converts electrical energy into

- a) Light energy
- b) Mechanical energy
- c) Heat energy
- d) Sound energy

Answer: c) Heat energy

67. Heating effect depends on

- a) Colour of wire
- b) Length of wire
- c) Current, resistance and time
- d) Shape of wire

Answer: c) Current, resistance and time

68. An electric fuse works on

- a) Magnetic effect
- b) Heating effect
- c) Chemical effect
- d) Optical effect

Answer: b) Heating effect

69. MCB stands for

- a) Main Circuit Board
- b) Miniature Circuit Breaker
- c) Magnetic Control Box

d) Maximum Current Breaker

Answer: b) Miniature Circuit Breaker

70.MCB automatically switches off during

a) Normal conditions

b) Power saving

c) Overload or short circuit

d) Lighting

Answer: c) Overload or short circuit

71.Magnetic effect of current was discovered by

a) Edison

b) Volta

c) Oersted

d) Faraday

Answer: c) Oersted

72.A current-carrying wire behaves like a

a) Heater

b) Magnet

c) Resistor

d) Battery

Answer: b) Magnet

73.An electromagnet loses magnetism when

a) Current flows

b) Battery is connected

c) Current stops

d) Wire heats

Answer: c) Current stops

74.The polarity of an electromagnet depends on

- a) Length of wire
- b) Thickness of nail
- c) Direction of current
- d) Size of battery

Answer: c) Direction of current

75.Electromagnets are used in

- a) Electric bells
- b) Mirrors
- c) Candles
- d) Thermometers

Answer: a) Electric bells

76.In a telephone, sound is produced by

- a) Heating effect
- b) Chemical effect
- c) Vibrating diaphragm
- d) Light effect

Answer: c) Vibrating diaphragm

77.The diaphragm in a telephone is made of

- a) Plastic
- b) Rubber
- c) Thin metal
- d) Wood

Answer: c) Thin metal

78.Chemical effect of electricity occurs in

- a) Metal wires

b) Conducting solutions

c) Insulators

d) Air

Answer: b) Conducting solutions

79. Electrolysis is the

a) Formation of ions

b) Heating of solution

c) Decomposition of molecules

d) Movement of electrons

Answer: c) Decomposition of molecules

80. Electroplating uses

a) Heat

b) Magnetism

c) Chemical effect

d) Light

Answer: c) Chemical effect

81. Electric current in a circuit flows from

a) Negative to positive terminal

b) Positive to negative terminal

c) Both directions

d) No direction

Answer: b) Positive to negative terminal

82. Resistance is similar to

a) Speed

b) Friction

c) Force

d) Pressure

Answer: b) Friction

83. A voltmeter is connected in

a) Series

b) Parallel

c) Closed loop

d) Open circuit

Answer: b) Parallel

84. Energy needed to move one unit charge defines

a) Current

b) Resistance

c) Potential difference

d) Conductivity

Answer: c) Potential difference

85. Materials with loosely bound electrons are

a) Insulators

b) Semiconductors

c) Conductors

d) Electrolytes

Answer: c) Conductors

86. Batteries have three main components including

a) Wire

b) Switch

c) Electrolyte

d) Bulb

Answer: c) Electrolyte

87. Dry cells are considered safe because their electrolyte is

- a) Solid
- b) Gas
- c) Paste
- d) Powder

Answer: c) Paste

88. Adding more bulbs in series causes

- a) Increase in brightness
- b) Decrease in brightness
- c) No change
- d) Explosion

Answer: b) Decrease in brightness

89. Adding more batteries increases

- a) Resistance
- b) Brightness
- c) Wire length
- d) Fuse melting

Answer: b) Brightness

90. Parallel circuits are preferred in homes because

- a) They are cheaper
- b) They save wire
- c) Appliances work independently
- d) They reduce voltage

Answer: c) Appliances work independently

91. Conductivity is highest in

- a) Insulators

b) Semiconductors

c) Conductors

d) Plastics

Answer: c) Conductors

92. Resistivity is lowest in

a) Rubber

b) Glass

c) Copper

d) Wood

Answer: c) Copper

93. Heating effect is useful in

a) Fans

b) Electric iron

c) Television

d) Radio

Answer: b) Electric iron

94. Electromagnets are temporary because

a) They are weak

b) They lose magnetism without current

c) They are small

d) They are costly

Answer: b) They lose magnetism without current

95. Chemical effect is used in

a) Motors

b) Generators

c) Electroplating

d) Bulbs

Answer: c) Electroplating

96. Electric energy is converted into sound in

a) Heater

b) Fan

c) Telephone

d) Iron

Answer: c) Telephone

97. Insulators are also called

a) Good conductors

b) Bad conductors

c) Semi conductors

d) Super conductors

Answer: b) Bad conductors

98. The flow of electrons constitutes

a) Heat

b) Light

c) Electric current

d) Voltage

Answer: c) Electric current

99. A closed circuit allows

a) No current

b) Partial current

c) Continuous current

d) Random current

Answer: c) Continuous current

100. Electricity today is considered a

- a) Luxury
- b) Rare resource
- c) Common household commodity
- d) Natural disaster

Answer: c) Common household commodity